

Name: _____ Date: _____

Period: _____

1. Factor the polynomial $p(x) = x^3 - x^2 - 6x$ completely.

$$p(x) = x(x - 3)(x + 2)$$

Zeros of p : $x = 0$, $x = 3$, $x = -2$

2. A rational function is given: $r(x) = \frac{x^2 - 9}{(x - 3)(x + 5)}$.

(a) Does r have a hole, a vertical asymptote, or both at $x = 3$? Circle: **Hole** / **Vertical Asymptote** / **Both**

(b) Identify any vertical asymptote(s). $x = -5$

3. The rational function $h(x) = \frac{3x^4 - x + 7}{2x^2 + 1}$ is written in standard form. Describe the end behavior of $h(x)$ as $x \rightarrow +\infty$.

End behavior: as $x \rightarrow +\infty$, $h(x) \rightarrow +\infty$

Teacher Note

Item 1: GCF first, then factor the trinomial. Zeros are 0, 3, -2.

Item 2(a): The numerator factors as $(x - 3)(x + 3)$, sharing $(x - 3)$ with the denominator. Since numerator multiplicity equals denominator multiplicity, this is a hole at $x = 3$.

Item 2(b): The only remaining denominator zero is $x = -5$ (not shared), so there is a VA there.

Item 3: Leading term of numerator is $3x^4$, denominator is $2x^2$, ratio is $\frac{3}{2}x^2 \rightarrow +\infty$.